Measurement Error and its Implications

Introduction to ME

ME stems from imperfect measurement processes.

It is everywhere but it is not a big problem when it is small and non-systematic.

Surveys are deeply prone to ME.

Introduction to ME

Classical ME:

$$X^* = X + U$$

Properties:

$$\begin{cases} E(U) = 0 \quad ; & \textit{null expectancy} \\ Var(U_i) = Var(U) \quad ; & \textit{homoskedasticity} \\ Cov(X, U) = 0 \quad ; & \textit{indep. error and true value} \\ Cov(u_i, u_j) = 0 \quad ; & \textit{no autocorrelation} \\ Cov(\in, U) = 0 \quad ; & non-differentiality \end{cases}$$

Implications of ME

Classical ME does not cause asymptotic bias in the sample mean.

$$X^* = X + 0$$

Although the level of precision will be affected.

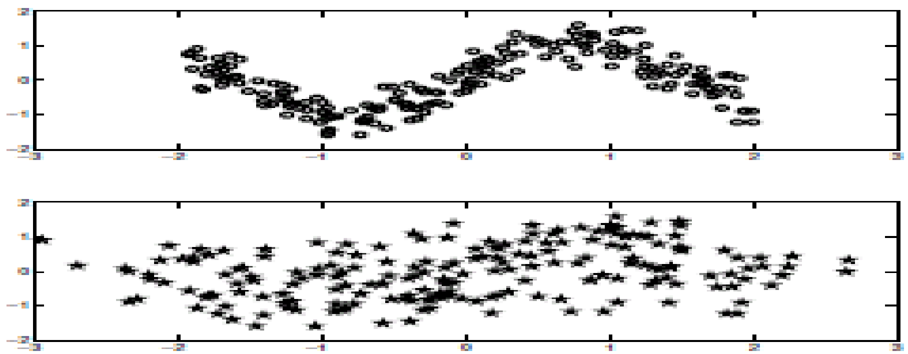
$$\sigma_{X^*}^2 = \sigma_X^2 + \sigma_{\mathrm{U}}^2$$

The more serious implications arise when using variables prone to ME in regression models.

Implications of ME

The Triple Whammy of ME

- ME masks relationships between variables



- It leads to a loss of power
- It bias the estimates of the regression coefficients

Implications of ME

<u>1st Misconception</u>: random ME does not bias the regression coefficients.

Simple linear regression,

$$Y = \beta_0 + \beta_1 X_1^* + \in$$

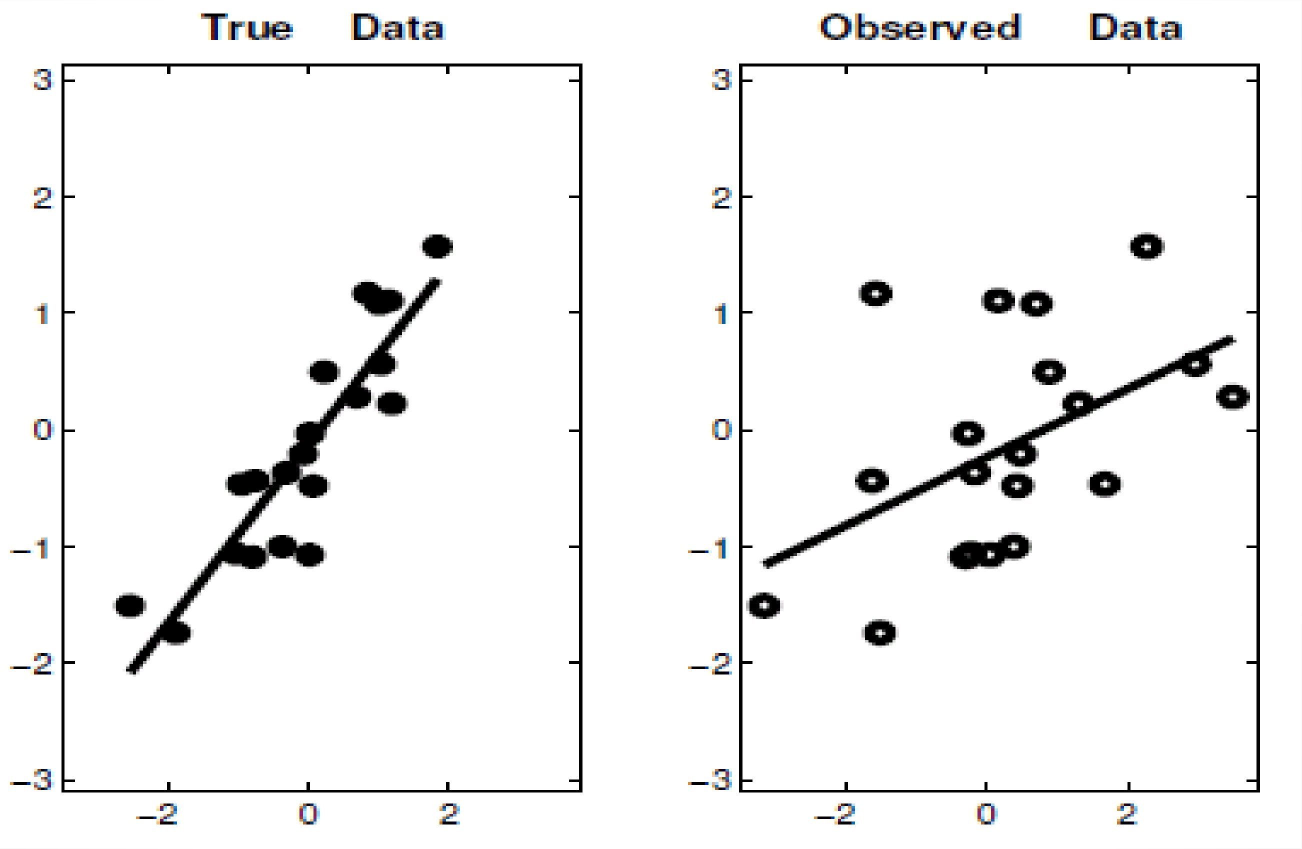
OLS estimator for the slope,

$$\beta_1 = \frac{\rho_{X,Y}}{\sigma_X^2}$$

Effect of classical ME,

$$\frac{\rho_{X^*,Y}}{\sigma_{X^*}^2} = \frac{\rho_{X,Y}}{\sigma_X^2 + \sigma_{\mathrm{U}}^2} = \frac{\rho_{X,Y}}{\sigma_X^2} \frac{\sigma_X^2}{\sigma_{X^*}^2} = \beta_1 \left(\frac{\sigma_X^2}{\sigma_X^2 + \sigma_U^2} \right)$$

Implications of ME



Source: Carroll et al, 2007

Implications of ME

2nd Misconception: ME only causes attenuation bias, the safest type of bias.

Simple linear regression,

$$\mathrm{Y} = \beta_0 + \beta_1 X^* + \beta_2 \mathrm{Z} + \epsilon$$

Effect on β_1,

$$\beta_{1^*} = \rho'_{X^*}\beta_1 = \frac{\sigma^2_{X|Z}}{\sigma^2_{X^*|Z}}\beta_1 = \frac{\sigma^2_{X|Z}}{\sigma^2_{X|Z} + \sigma^2_U}\beta_1$$

Effect on β_2,

$$\beta_{2^*} = \beta_2 + \beta_1(1 - \rho'_{X^*})\Gamma_Z$$

Implications of ME

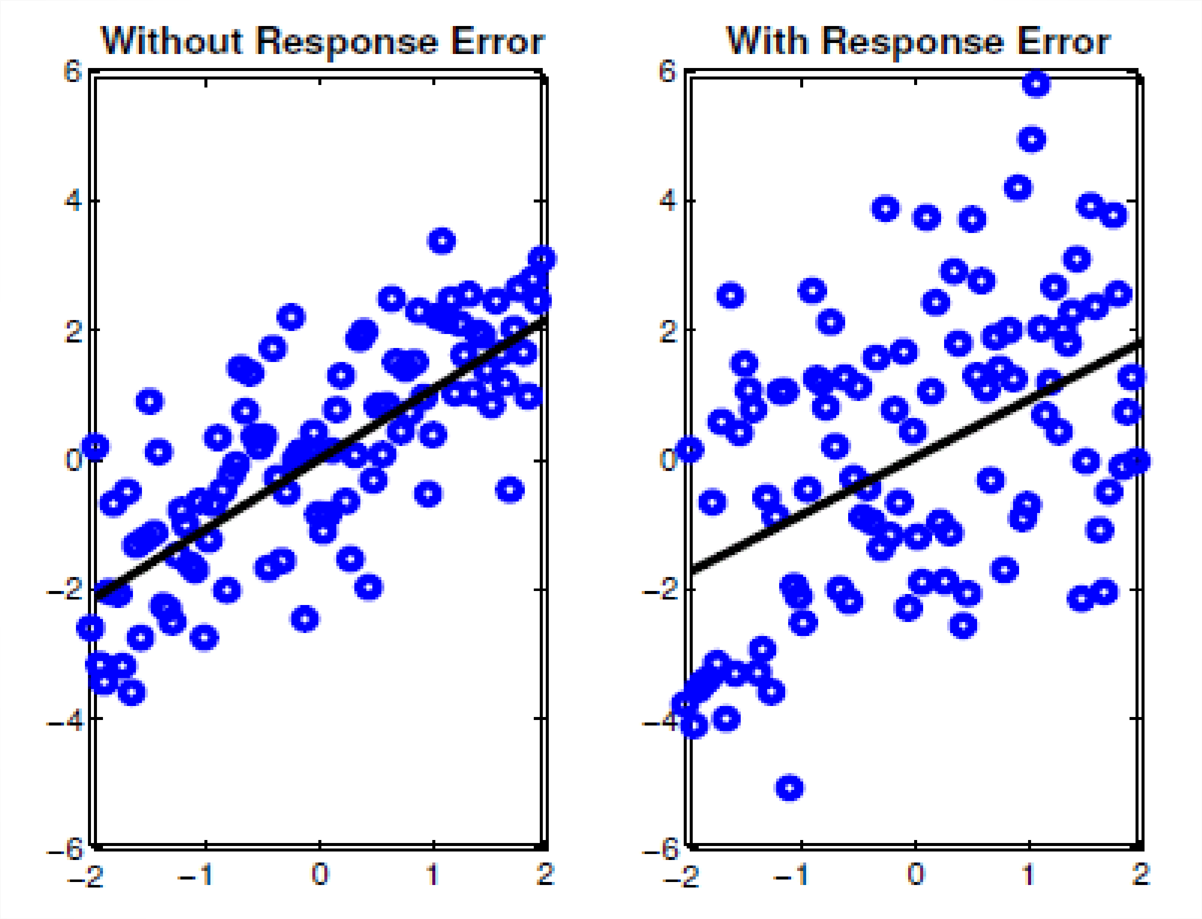
3rd <u>Misconception</u>: ME in the response variable has an effect on precision but does not bias regression coefficients.

$$Y^* = Y + U$$

Simple regression with classical ME in the response,

$$Y^* = \beta_0 + \beta_1 X + (\epsilon + U)$$

Implications of ME



Source: Carroll et al, 2007

Implications of ME

However, bias will normally appear if the ME departs from the classical ME model.

Additive ME makes no sense in categorical variables. Instead the ME model is defined by the misclassification matrix.

$$\theta_{Y^*|Y} = P(Y^* = y^*|Y = y)$$

For the case of a binary variable,

$$\begin{cases} P(y^* = 1|y = 1) = \theta_{1|1}\ ; & \quad Sensitivity \\ P(y^* = 0|y = 0) = \theta_{0|0}\ ; & \quad Specificity \end{cases}$$

$$\begin{cases} P(y^* = 0|y = 1) = \theta_{0|1}\ ; & \quad Probability\ of\ a\ false\ negative \\ P(y^* = 1|y = 0) = \theta_{1|0}\ ; & \quad Probability\ of\ a\ false\ positive \end{cases}$$

Implications of ME

For the specification of a binary response variable, the linear regression is now adapted with a link function,

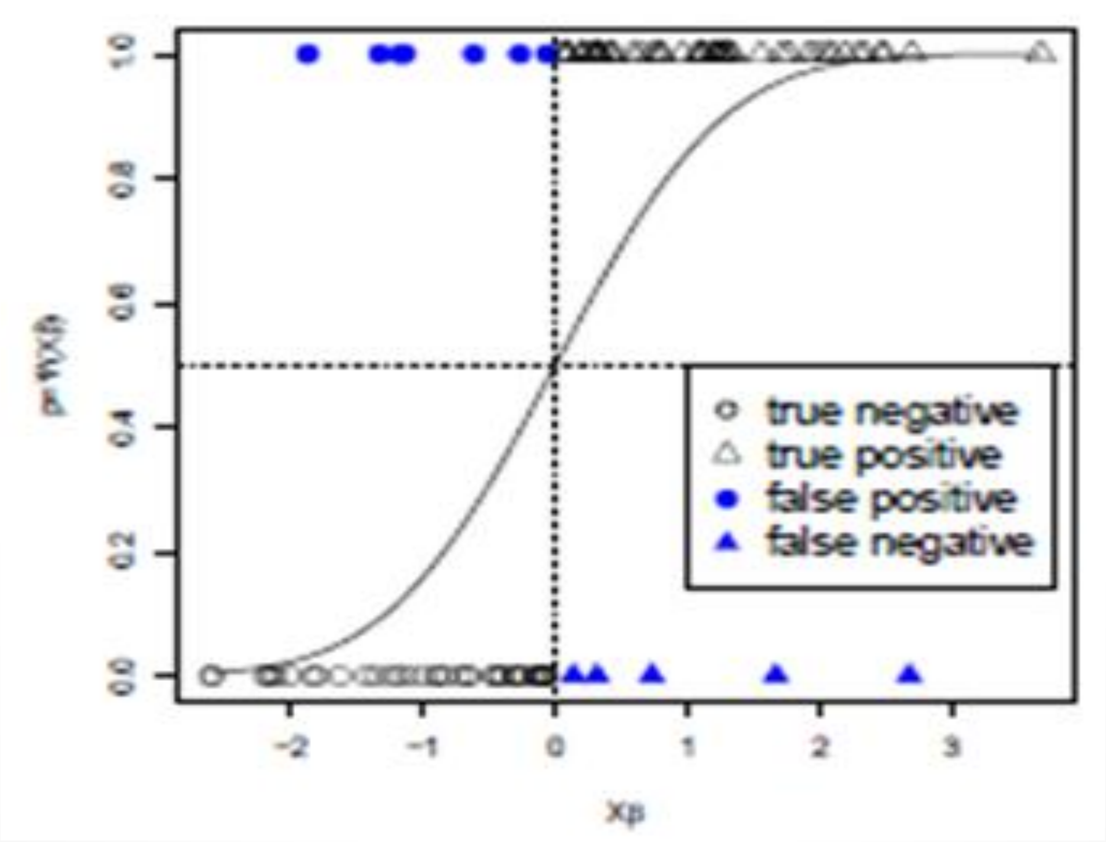
$$P(Y = 1|X) = h(\beta_0 + \beta_1 X)$$

However, given that we observe Y^* with misclassification matrix $\theta_{Y^*|Y}$, the model that we are really estimating is

$$P(Y^* = 1|X) = \theta_{1|1} P(Y = 1|X) + (1 - \theta_{0|0})(1 - P(Y = 1|X))$$

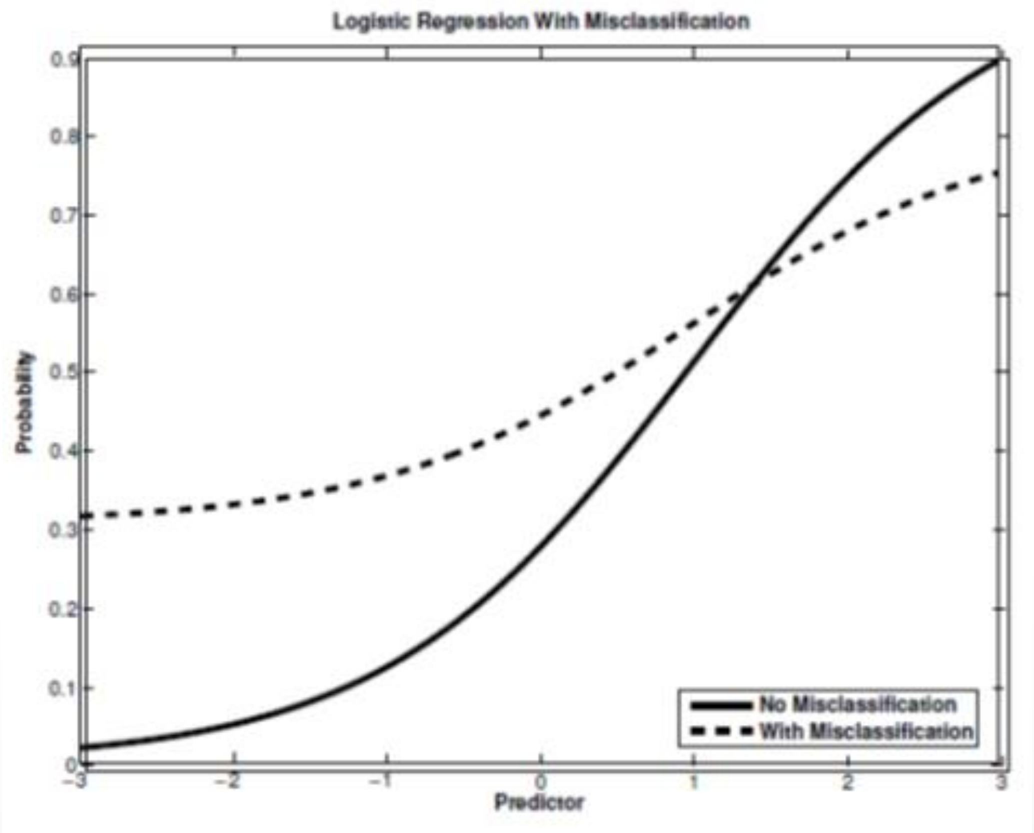
Implications of ME

Non-differential misclassification attenuates the slope



Implications of ME

The magnitude of the attenuation is given by the level of sensitivity, specificity and the size and sign of the intercept.



Implications of ME

In summary, the effect of ME is difficult to anticipate.

Some important factors that will determine its impact and traceability are:

The type of measurement error (classical, misclassification, etc.),

The outcome model ,

Correlations of the ME with other predictors and with the true value.

"Measurement error is, to borrow a metaphor, a gremlin hiding in the details of our research that can contaminate the entire set of estimated regression parameters"

(Nugent, et al. 2000, pp.60)

ME in Retrospective Questions on Work-Status

I look at ME in retrospective questions on work-status (1st part of my Phd).

I can assess the presence of ME thanks to a validation dataset from a register of unemployment.

ME from a count data perspective,

Time-frame	% of subjects reporting spells correctly	spells reported as a % of spells registered
1 year	54.00%	86.30%
12 years	7.50%	38.80%

ME in Retrospective Questions on Work-Status

ME from a duration data perspective,

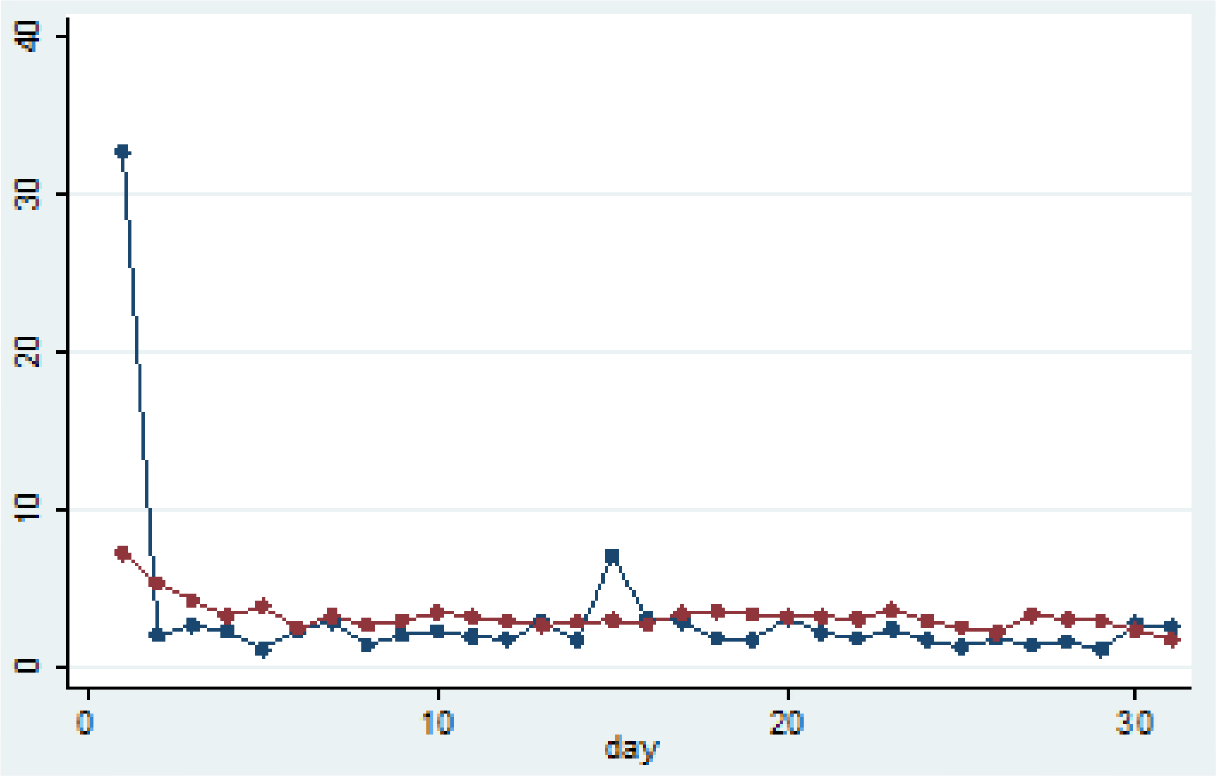
Time-frame	Reported duration time for spells of unemployment as the % of the registered time
1 year	109%
12 years	52%

ME from a person-period perspective,

Time-frame	Percentage of matched cases
1 year	68.4
12 years	62.1

ME in Retrospective Questions on Work-Status

Misdating (heaping effects),

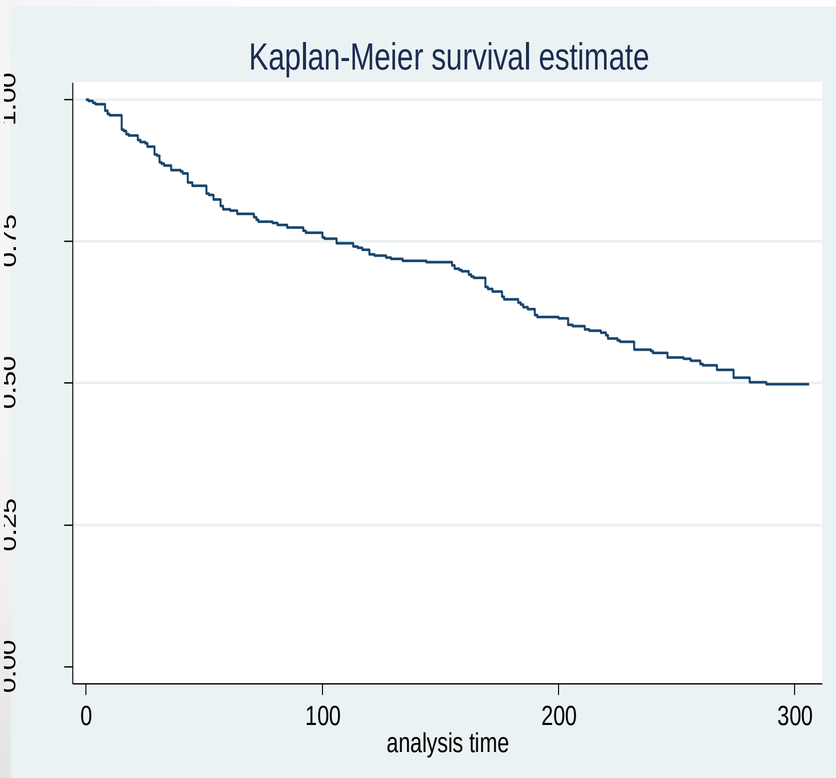


ME in Event History Analysis

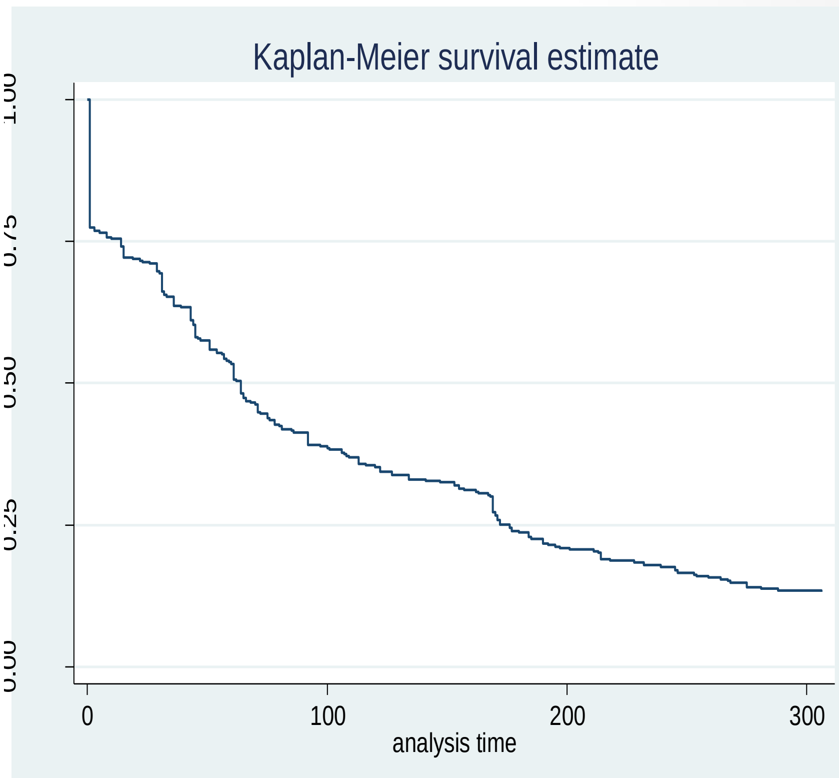
I assess the impact of ME on EHA (2[nd] part of my Phd).

Impact on descriptive statistics,

Register



Survey



ME in Event History Analysis

Impact on the estimates of regression parameters,

Register

	Regression Coefficients	Standard Errors	P-values
Age	0.023	0.01	0.017
Work time	0.47	0.338	0.164
Constant	4.476	0.503	0
LR chi2(2)	8.29		0.016
Subjects	303		

Survey

	Regression Coefficients	Standard Errors	P-values
Age	0.021	0.012	0.087
Work time	-0.664	0.318	0.037
Constant	4.515	0.51	0
LR chi2(2)	5.85		0.053
Subjects	303		

Methods for the Adjustment of ME

I compare the effectiveness of methods for the adjustment of ME (3rd part of my Phd).

Methods that require additional variables :

- Factor analysis (LISREL models)
- Instrumental variables

Methods that require replicated or validation data:

- Regression calibration
- Corrected scores

Methods that only require knowledge (assumptions) about the behavior of the ME :

- Maximum likelihood-based methods
- Simulation-extrapolation (SIMEX)